

Simple Python SIEM - Documentation

<https://github.com/elvistang199/Python-SIEM>

Overview

This document explains the structure and functionality of a SIEM (Security Information and Event Management) system written in Python. It simulates log generation, real-time monitoring, parsing, storing, and alerting.

Project Structure

The project folder contains the following files and directories:

```
simple_siem/
├── main.py           # Entry point: runs generator and monitor
├── collector.py      # Simulates log collection
├── generator.py      # Generates logs in real-time (simulated)
├── monitor.py        # Monitors the log file for new events
├── parser.py         # Parses raw logs into structured format
├── storage.py        # Stores logs into a local SQLite database
├── alert.py          # Detects alert-worthy events
├── logs/
│   └── raw_logs.txt  # File containing generated raw logs
├── backend/
│   └── siem.db       # SQLite database storing structured logs
```

main.py

This file serves as the main entry point of the SIEM. It creates two threads: one to simulate log generation and another to monitor the log file in real-time.

Key Functions:

- - Starts log generator from generator.py
- - Starts file monitor from monitor.py
- - Uses an infinite loop to keep the program running

generator.py

This module simulates real-time log generation. It writes logs to 'logs/raw_logs.txt' every 5 seconds, mimicking login and file access events.

monitor.py

This module continuously monitors 'logs/raw_logs.txt' for new lines. When a new log entry is added, it parses, stores, and checks for alerts.

parser.py

Parses each log line into a structured format (dictionary) with three fields:

- - timestamp
- - event_type
- - description

storage.py

Initializes and connects to a SQLite database located at 'backend/siem.db'. Each parsed log is inserted into the 'logs' table for storage.

alert.py

Checks logs for specific patterns and conditions. For example, it alerts if two or more 'LOGIN_FAIL' events occur.

How to Run the SIEM

1. Open a terminal in the project folder
2. Run the program using:

python main.py

3. Logs will be generated and monitored in real-time

How to Stop the SIEM

To stop the SIEM, press Ctrl+C in the terminal. This will interrupt the infinite loop and gracefully exit the program.